Supervised by:
Dr. Mohammed Sharkawy

Nephrology
2026

Out of the League
Family Medicine

Prepared by: Dr. Haya Alturkait

3rd Edition

Table of Contents

Topic: Autosomal dominant polycystic kidney disease (ADPKD) 2
Topic: Alport's syndrome 4
Topic: Acute kidney injury 5
Topic: Chronic Kidney Disease 13
Topic: Chronic kidney disease and anemia 26
Topic: Chronic kidney disease and mineral bone disease management 27
Topic: Hematuria 28
Topic: IgA vasculitis (Henoch-Schönlein purpura) 31
Topic: IgA nephropathy (IgAN) 32
Topic: Post-streptococcal glomerulonephritis (PSGN) 33
Topic: Minimal change disease (MCD) 34
Topic: Renal papillary necrosis 36
Topic: Diabetes insipidus 37
Topic: Spironolactone 38

Topic: Autosomal dominant polycystic kidney disease (ADPKD)

Autosomal dominant polycystic kidney disease (ADPKD) is the most common inherited cause of kidney disease, affecting 1 in 1,000 Caucasians. Two disease loci have been identified, PKD1 and PKD2, which code for polycystin-1 and polycystin-2 respectively.

Features:

Patients with ADPKD may be asymptomatic or present with:

- Flank pain, due to kidney hemorrhage, obstructive calculi, or urinary tract infection, is the most common symptom reported by patients.
- Hypertension.
- Proteinuria
- Recurrent UTIs.
- Flank pain.
- Hematuria.
- Palpable kidneys.
- Renal impairment.
- Renal stones.

Extra-renal manifestations:

- Liver cysts (70% - the commonest extra-renal manifestation): may cause hepatomegaly
- Berry aneurysms (8%): rupture can cause subarachnoid hemorrhage
- Cardiovascular system: mitral valve prolapse, mitral/tricuspid incompetence, aortic root dilation, aortic dissection
- Cysts in other organs: pancreas, spleen; very rarely: thyroid, esophagus, ovary

Family History:

Data important to elicit on family history include:

- Number and relationship of family members affected
- Age at diagnosis
- Their age of developing end-stage kidney disease (ESKD; if applicable)
- Any known genetic mutations in the family.

Diagnosis:

The **initial step** toward diagnosis of ADPKD is to obtain a detailed family history from the patient. The diagnosis of ADPKD is confirmed primarily by imaging.

Investigations:

- The screening investigation for relatives is **abdominal ultrasound**.
- Among patients with a family history of ADPKD, who are asymptomatic with normal kidney function, an **ultrasound** is usually sufficient to diagnose or rule-out the presence of disease.
- (MRI) may be necessary if the results of the ultrasound are equivocal or to further evaluate incidentally detected complications (e.g., kidney mass or complex cysts).
- RFT, eGFR.

Management:

Explanation:

- Polycystic kidney disease (PKD) is a condition that affects the kidneys.
- When people have PKD, abnormal fluid-filled sacs called "cysts" grow in the kidneys.
- The cysts cause the kidneys to get bigger than normal.
- The cysts can also keep the kidneys from working normally.
- This can lead to problems, such as high blood pressure, kidney infections, and kidney failure. Kidney failure is when the kidneys stop working completely.
- Besides kidney problems, PKD can cause problems in other parts of the body.
- PKD usually runs in families.

Advices:

- BP control: Target BP is less than 110/75.
- Advise all ADPKD patients to **restrict dietary sodium**, with a goal of 2 grams intake per day or less. In addition, advise patients with ADPKD to **drink >3 L of fluid** per day, unless the eGFR is <30 mL/min/1.73 m2 or the patient is at risk for hyponatremia.
- Patients who are overweight (a body mass index [BMI] of 25 to 30) and patients with obesity (BMI greater than 30) have faster progression of kidney disease in ADPKD. Thus, up-to-date recommend keeping a healthy BMI (<25)
- There are conflicting findings on the benefit of a low-protein diet in people with ADPKD. Given the limited evidence of benefit, UpToDate do not recommend restricting protein intake below 0.8 g/kg of body weight per day.

Prescription:

For BP: ADPKD usually joins bilateral renal artery stenosis and nephrosclerosis in which ACE-I may deteriorate renal function.

For select patients, **tolvaptan** (vasopressin receptor 2 antagonist) may be an option.

Tolvaptan is a type of drug called a vasopressin receptor antagonist. It has been shown to slow the decline in kidney function and reduce pain symptoms in people with ADPKD.

NICE Indications for using **tolvaptan**:

1. Chronic kidney disease stage 2 or 3 at the start of treatment
2. Evidence of rapidly progressing disease

Referral:

- Patients with ADPKD who progress to ESKD require kidney replacement therapy.
- Peritoneal dialysis is less commonly performed than hemodialysis because it is perceived to be difficult for the patients to accommodate large volumes of peritoneal dialysate fluid in the setting of massively enlarged kidneys.
- Patients with ADPKD and ESKD are therefore most commonly treated with hemodialysis or undergo kidney transplantation.

Screening:

- Screening tests are available to relatives of a person with autosomal dominant polycystic kidney disease (ADPKD)

Topic: Alport's syndrome

Alport's syndrome is usually inherited in an X-linked dominant pattern*.
It is due to a defect in the gene which codes for type IV collagen resulting in an abnormal glomerular-basement membrane (GBM). The disease is more severe in males with females rarely developing renal failure.

Alport's syndrome usually presents in childhood. The following features may be seen:

- Asymptomatic microscopic hematuria. Gross hematuria may also be a presenting finding, which may occur after an upper respiratory infection. Glomerular disease that progresses to kidney failure
- Bilateral sensorineural deafness
- Lenticonus: protrusion of the lens surface into the anterior chamber
- Ocular abnormalities (e.g., anterior lenticonus, Retinitis pigmentosa)
- Renal biopsy: splitting of lamina densa seen on electron microscopy.

Investigations:

- RFT
- Urine protein
- Renin-angiotensin (RAS) to be done Annually, (if the patient has Alport + proteinuria, and takes ACE inhibitors or ARBS).

Diagnosis:
Pass Medicine:

1. Molecular genetic testing.
2. Renal biopsy:
    - Electron microscopy: characteristic finding is of the longitudinal splitting of the lamina densa of the glomerular basement membrane, resulting in a 'basket-weave' appearance:

Referral:

- Nephrology, OPD
- Supportive measures are used for hearing loss (eg, hearing aids)
- Ocular impairment in affected patients.

Treatment:

- There is no specific treatment to correct the underlying defect for Alport syndrome.
- ACE or ARBs in patients with Alport disease and overt proteinuria (reduces the rate of protein excretion and appears to slow the progression of the disease)
- Cyclosporine therapy it’s not suggested in patients with Alport syndrome (UpToDate)
- Either dialysis or transplantation can be performed in patients with Alport syndrome who develop kidney failure (stage 5 chronic kidney disease).

Topic: Acute kidney injury

Definition:

- Acute kidney injury (AKI) is a term covering a spectrum of injury to the kidneys that can result from a number of causes which may co-exist. It is a clinical syndrome rather than a biochemical diagnosis.
    - It is characterized by a sudden decline in renal excretory function over hours or days that can result in failure to maintain fluid, electrolyte, and acid-base balance. The KDIGO guideline defined AKI as an abrupt decrease in kidney function occurring over 7 days or fewer.
    - The diagnosis of AKI and its staging is based on acute changes in serum creatinine and/or a reduction in urine output (oliguria).
    - The term 'acute kidney injury' has replaced the concept of 'acute renal failure', as it more accurately describes a wide range of injury (structural damage) and impairment (loss of function) causing earlier stages of kidney damage, not just kidney failure.

Causes:

- There are multiple possible causes of acute kidney injury (AKI) which may co-exist, which have traditionally been divided into:
    - Pre-renal (most common) — due to reduced perfusion of the kidneys and/or hypotension leading to a decreased glomerular filtration rate (GFR). It is usually reversible with appropriate early treatment.
    - Intra-renal — a consequence of structural damage to the kidney, for example, tubules, glomeruli, interstitium, and intrarenal blood vessels. It may result from persistent pre-renal or post-renal causes damaging renal cells.
    - Post-renal (least common, accounting for around 10% of AKI) — due to acute obstruction of urine flow within the renal tract resulting in increased intratubular pressure and decreased GFR.
- More than one contributing cause is often present

Table 1. Possible causes of acute kidney injury (AKI).

Pre-renal	Renal	Post-renal
Hypovolaemia (for example inability to maintain hydration without help from others, haemorrhage, gastrointestinal losses, renal losses, and burns)	Toxins and nephrotoxic drugs (for example antibiotics, nonsteroidal anti-inflammatory drugs, proton pump inhibitors, allopurinol, iodine-based contrast, and chemotherapy)	Obstruction (for example. renal or ureteric stones, blocked catheter, enlarged prostate, abdominal or pelvic mass or tumour, retroperitoneal fibrosis, pyonephrosis, previous hydronephrosis; recurrent urinary tract infections, and neurogenic bladder)

Reduced cardiac output leading to hypotension (for example heart failure, liver failure, sepsis, and drugs)	Vascular (for example vasculitis, thrombosis, athero/thromboembolism, and arterial dissection)	
Drugs which reduce blood pressure, circulating volume, or renal blood flow (for example ACE inhibitors*, ARBs**, NSAIDs†, loop diuretics, and mineralocorticoid receptor antagonists such as spironolactone)	Glomerular (for example glomerulonephritis)	
	Tubular (for example acute tubular necrosis, rhabdomyolysis, and myeloma)	
	Interstitial (for example interstitial nephritis, and lymphoma infiltration)	
* Angiotensin-converting enzyme inhibitor **Angiotensin receptor blocker † Nonsteroidal anti-inflammatory drug		

Risk factors:

The risk of acute kidney injury (AKI) depends on a person's susceptibility and the type and extent of exposure to a potential insult. Those at increased risk of AKI include people:

- Aged 65 years or over.
- With a history of AKI.
- With Chronic kidney disease (people with an estimated glomerular filtration rate [eGFR] less than 60 mL/min/1.73m2 are at particular risk).
- With symptoms or history of urological obstruction or conditions which may lead to obstruction.
- With chronic conditions such as heart failure, liver disease, and diabetes mellitus (particularly if proteinuria).
- With neurological or cognitive impairment or disability (may be reliant on others for fluid intake and at risk of reduced intake).
- With Sepsis.
- With hypovolaemia; hypotension; dehydration; or reduced fluid intake.
- With oliguria (urine output less than 0.5 mL/kg/hour).

- Who have used specific medication within the last week (especially if hypovolaemic) — such as nonsteroidal anti-inflammatory drugs (NSAIDs), angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists (ARBs), diuretics including mineralocorticoid receptor antagonists, and antibiotics including aminoglycosides.
- Exposed to iodine-based contrast agents within the past week.
- With cancer and or undergoing cancer therapy (risk will depend on the type of cancer, proposed treatment, and premorbid risk factors).
- Who are immunocompromised (such as untreated HIV infection or malnourishment) or immunosuppressed.
- Exposed to toxins (for example some herbal remedies and poisonous plants and animals).

Complications:

- Complications from acute kidney injury (AKI) occur as a result of impairment of the kidney's excretory, endocrine, and metabolic actions. The risk of complications is related to the severity and duration of AKI, the clinical context, and the person's comorbidities, and can involve multiple organ systems, including:
    - Hyperkalaemia.
        - This is usually asymptomatic until severe, and may cause muscle weakness, paralysis, cardiac arrhythmias, or in extreme cases cardiac arrest.
    - Other electrolyte imbalances, for example, hyperphosphataemia, hypernatremia or hyponatraemia, hypomagnesaemia or hypermagnesaemia, and hypocalcaemia.
    - Metabolic acidosis.
        - Can present with altered level of consciousness, circulatory collapse, and hyperventilation.
    - Fluid overload (peripheral and pulmonary oedema); heart failure; lung injury; and liver injury.
        - Signs include tachypnoea, tachycardia, cyanosis, and lung crepitations.
    - Uraemia.
        - Occurs in severe AKI, and may be associated with pericarditis, neuropathy, or encephalopathy requiring renal replacement therapy.
        - Symptoms include confusion, lethargy, and altered level of consciousness.
    - Chronic kidney disease (CKD), progressive loss of renal function over time, and end-stage kidney disease.

Prognosis:

Evidence from epidemiological studies suggests that acute kidney injury (AKI) has important clinical consequences, even in mild, reversible cases. It has potentially variable outcomes, with increasing severity of AKI correlating with a higher risk of a worse prognosis.

- AKI can contribute to increased length of hospital stay, together with higher in-hospital and longer-term mortality rates.

Diagnosis:

Acute kidney injury (AKI) is often asymptomatic, especially in the early stages.

- Suspect a diagnosis of AKI in a person with:
  - Symptoms and signs such as:
    - Nausea and vomiting, diarrhoea, or suspected dehydration.
    - Reduced urine output or changes to urine colour.
    - Confusion, fatigue, or drowsiness.
  - An acute illness and any risk factors for AKI.
  - An illness with no clear acute component and any of the following:
    - Chronic kidney disease (CKD, especially stage 3B, 4, or 5), or urological disease.
    - New-onset or significant worsening of urological symptoms.
    - Symptoms suggesting a complication of AKI.
    - Symptoms or signs of a multisystem disease affecting the kidneys and other organ systems (for example thrombotic thrombocytopenic purpura).
  - An AKI warning stage test result generated from electronic detection systems in a biochemistry laboratory. This flags up changes in creatinine levels suggestive of AKI in primary care.

An acute kidney injury (AKI) warning stage test result may be generated from electronic detection systems in a biochemistry laboratory. This flags up changes in creatinine levels suggestive of AKI in primary care.

- Respond to AKI warning stage test results within an appropriate timescale using clinical judgement, as suggested below. Consider arranging an urgent review and clinical assessment if there are:
  - Poor fluid intake or urine output.
  - Evidence of hyperkalaemia (especially if moderate with potassium 6–6.4 mmol/L or severe with potassium 6.5 mmol/L or more).
  - Previous AKI.
  - Known chronic kidney disease (CKD) stage 4 or 5 or history of renal transplant — urgent liaison with a nephrologist may be needed.
  - A history of frailty and/or chronic disease such as diabetes, liver disease, or heart failure.
  - Suspected intrinsic kidney disease or urinary tract obstruction.
- If there is a creatinine change suggesting a potential AKI, take into account the clinical context (for example assessment of acute illness, chronic disease monitoring, or drug monitoring).

- Take into consideration if the person has:
  - Chronic kidney disease (CKD) — an increase in serum creatinine may be due to progression of CKD rather than AKI. Assess the pattern of serum creatinine values over a longer period of time, and consider arranging a repeat serum creatinine measurement to see if changes are sudden or stable, depending on clinical judgement.
  - Recently been treated with trimethoprim — this can cause a false positive result, as trimethoprim may increase serum creatinine, but not affect the GFR.
  - Recently completed a pregnancy — this can cause a false positive result due to an apparent rise in creatinine compared with naturally reduced creatinine values in pregnancy.

Assessment:

- If a person has confirmed acute kidney injury (AKI), assess:
  - Volume status and possible underlying cause(s) by checking:
    - Fluid intake and losses.
    - Pulse/blood pressure (including any postural changes) and jugular venous pressure — particularly for signs of hypotension and hypovolaemia.
    - Peripheral perfusion (capillary refill time); moistness of mucous membranes; and skin turgor — for signs of dehydration.
    - Changes in urinary output and when last passed urine.
    - For peripheral oedema and crackles on lung auscultation — for signs of fluid overload.
    - For a palpable bladder, abdominal or pelvic mass, or enlarged prostate (if clinically appropriate) — for signs of urinary tract obstruction.
  - Renal function and serum potassium level — to assess for hyperkalaemia.
  - For possible underlying cause and the likelihood of AKI, by asking about:
    - Current symptoms, including acute intercurrent illness (such as diarrhoea and vomiting, urinary or renal tract infection, and sepsis) and symptoms suggesting urinary tract obstruction (for example lower urinary tract symptoms, bloating from a pelvic mass, and renal colic).
    - Fluid intake and any reduction in urine output (oliguria or anuria).
    - Any co-morbidities or risk factors for AKI.
    - Any systemic symptoms associated with intrinsic renal disease (for example vasculitic rash, arthralgia, uveitis, epistaxis, or haemoptysis).
    - Drug history — any medication that can cause or exacerbate AKI, or that may accumulate and cause harm during AKI; and any recent dose increases, including over-the-counter preparations, herbal remedies, and recreational drugs such as ketamine, synthetic cannabinoids, and 'legal highs'.

        - Risk factors for rhabdomyolysis, such as skeletal muscle injury, muscle overexertion, crush injury, or prolonged immobility.
        - Social situation and need for a carer to support fluid intake.
    - For renal disease by urine dipstick analysis for blood, protein, leucocytes, nitrites, and glucose.
        - AKI with negative urinalysis usually indicates a pre-renal or drug cause.
        - Positive protein and blood may suggest glomerular disease (particularly if 2+ blood, 2+ protein) if there is no evidence of urinary tract infection or trauma due to catheterisation.
        - Increased white cells are non-specific but may suggest infection (most common) or interstitial nephritis.
        - Note: be aware that urine dipstick analysis in a person with a catheter in situ should be interpreted with caution as there may be false-positive results (such as haematuria due to simple trauma).
    - The stage of AKI.
        - Assess the stage of AKI according to the criterion which gives the highest (worst) stage. See Table 2 for more detailed information.
        - In a primary care setting, the creatinine level will usually be the most readily available result, but if urine output is also being monitored, base staging on whichever shows the highest (worst) stage if creatinine and urine output correspond with different stages.

Table 2. Summarized staging system for AKI in adults (based on the KDIGO§ classification).

Stage	Criteria
1	Creatinine rise of 26 micromol or more within 48 hours or Creatinine rise of 50–99% from baseline within 7 days* (1.50–1.99 x baseline) or Urine output** less than 0.5 mL/kg/hour for more than 6 hours
2	100–199% creatinine rise from baseline within 7 days* (2.00–2.99 x baseline) or Urine output** less than 0.5 mL/kg/hour for more than 12 hours
3	200% or more creatinine rise from baseline within 7 days* (3.00 or more x baseline) or Creatinine rise to 354 micromol/L or more with acute rise of 26 micromol/L or more within 48 hours or 50% or more rise within 7 days or Urine output** less than 0.3 mL/kg/hour for 24 hours or anuria for 12 hours

* The rise is known (based on previous blood tests) or presumed (based on history) to have occurred within 7 days.

** Measurement of urine output may not be practical in primary care, but can be considered in a person with a catheter.

Management:

From age 18 years onwards.
How should I manage a person with acute kidney injury?

Take into account factors such as the stage of acute kidney injury (AKI), the person's age and comorbidities, risk of complications, and social situation when assessing whether to arrange hospital admission, liaison with a specialist, or ongoing management in primary care.

- Arrange urgent hospital admission or same-day referral, depending on clinical judgement, if a person has:
  - Likely stage 3 AKI.
  - An underlying cause requiring urgent hospital management such as suspected urinary tract obstruction and/or upper urinary tract infection.
  - No identifiable cause for AKI.
  - A risk of urinary tract obstruction.
  - Sepsis.
  - Evidence of hypovolaemia and a need for intravenous fluid replacement and monitoring.
  - A deterioration in clinical condition or a need for observation or monitoring of a frequency that is impractical in primary care.
  - A suspected complication of AKI requiring urgent hospital management such as pulmonary oedema, uraemic encephalopathy, pericarditis, or severe hyperkalaemia (serum potassium of 6.5 mmol/L or more).
- Liaise with a nephrologist as soon as possible and within 24 hours of detection, if one or more of the following is present:
  - Stage 4 or 5 chronic kidney disease (CKD).
  - A possible diagnosis that may need specialist treatment, for example, tubulointerstitial nephritis, glomerulonephritis (indicated by haematuria and/or proteinuria on urine dipstick analysis), systemic vasculitis with renal involvement, or myeloma.
  - An inadequate response to treatment in primary care.
  - A history of renal transplant.

- For people with stage 1 AKI who do not need hospital admission, referral, or specialist input, provide management in primary care.
    - Manage the underlying cause, if the expertise and resources are available in primary care.
    - Offer supportive measures such as advice on maintaining fluid balance during intercurrent illness.
    - Consider temporarily stopping specific medications during intercurrent illness, or adjusting the doses of medication in relation to renal function, until the person's clinical condition has improved. Seek specialist advice if needed.
    - Monitor serum creatinine regularly, using clinical judgement to determine the frequency.
        - Be aware that even small increases in serum creatinine can be significant.
    - Reconsider the need to arrange hospital admission or liaison with a specialist if there is any deterioration in the person's clinical condition, or an inadequate response to treatment in primary care.

From age 18 years onwards.
How should I prevent acute kidney injury?

If a person has risk factors for developing acute kidney injury (AKI):

- Arrange regular monitoring of serum creatinine, basing the frequency on the person's individual risk, clinical context, and clinical judgement.
    - Arrange regular monitoring of renal function in people with chronic diseases including chronic kidney disease (CKD), heart failure, liver disease, and diabetes.
    - Arrange close monitoring of renal function if a person has an acute illness, especially if there is diarrhoea, vomiting and/or signs of dehydration.
- Explain the risk of developing AKI is increased with:
    - Acute illness and inability to maintain fluid intake (such as diarrhoea and vomiting), and/or
    - Specific medications that may need to be temporarily stopped during intercurrent illness (including over-the-counter preparations and herbal remedies) due to risk of hypovolaemia and/or nephrotoxicity. This is particularly important in people with:
        - CKD with an estimated glomerular filtration rate (eGFR) less than 60 mL/min/1.73 m2.
        - Neurological or cognitive impairment or disability (may be reliant on others for fluid intake and at risk of reduced intake).

Topic: Chronic Kidney Disease

Definition:

- Chronic kidney disease (CKD) is defined as a reduction in kidney function or structural damage (or both) present for more than 3 months, with associated health implications.
    - Kidney damage may cause fluid and electrolyte imbalance, and leakage of protein and/or blood into the urine, resulting in proteinuria and haematuria.
- CKD should be diagnosed in people with:
    - Markers of kidney damage such as a urinary albumin:creatinine ratio (ACR) greater than 30 mg/g, urine sediment abnormalities, electrolyte and other abnormalities due to tubular disorders, abnormalities detected by histology, structural abnormalities detected by imaging, and a history of kidney transplantation, and/or
        - Urine sediment abnormalities include red blood cells; white blood cells; cellular, fatty, or granular casts; or renal tubular epithelial cells.
    - A persistent reduction in renal function shown by a serum estimated glomerular filtration rate (eGFR) of less than 60 mL/min/1.73 m2.
        - The GFR is a measure of the total amount of fluid filtered through all the functioning kidney nephrons per unit of time, and is derived from serum creatinine using standard estimating equations.
- CKD is classified based on the underlying cause, GFR and proteinuria category.
    - 'Accelerated progression' of CKD is defined as a persistent decrease in eGFR of a persistent decrease in eGFR of 15 mL/min/1.73 m2 within 12 months.

Causes:

There are multiple possible causes and risk factors for chronic kidney disease (CKD) and its progression, including:

- Conditions associated with intrinsic kidney damage, such as:
    - Hypertension.
    - Diabetes mellitus.
    - Glomerular disease, such as acute glomerulonephritis (typically if CKD follows a Streptococcal upper respiratory tract infection; may also follow hepatitis B, hepatitis C, or HIV infection).
- Current or previous history of acute kidney injury (AKI).
    - This may be triggered by severe intercurrent illness or dehydration.
- Potentially nephrotoxic drugs, such as:
    - Aminoglycosides, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor antagonists (AIIRAs), bisphosphonates, calcineurin inhibitors (such as ciclosporin or tacrolimus), diuretics, lithium, mesalazine, and nonsteroidal anti-inflammatory drugs (NSAIDs).
- Conditions associated with obstructive uropathy, such as:

    - Structural renal tract disease.
    - Bladder voiding problems such as neurogenic bladder, benign prostatic hypertrophy.
    - Urinary diversion surgery.
    - Recurrent urinary tract calculi.
- Multisystem diseases with potential renal involvement, such as:
    - Systemic lupus erythematosus (SLE) which may cause lupus nephritis.
    - Vasculitis.
    - Myeloma.
- A family history of CKD stage 5, or hereditary kidney disease such as autosomal dominant polycystic kidney disease, Alport's syndrome (a sex-linked recessive disease which typically causes haematuria), and familial glomerulonephritis.
- Cardiovascular disease.
- Obesity with metabolic syndrome.
    - Obesity alone is not a risk factor for CKD.
- Gout.
- People with an incidental finding of haematuria or proteinuria.

For people with any of these risk factors offer testing for CKD using eGFR:creatinine and ACR and monitor at least annually for people on potentially nephrotoxic drugs.

Complications:

The risk of complications from chronic kidney disease (CKD) increases with disease progression. These may include:

- Acute kidney injury.
    - This may initiate or accelerate CKD progression.
- Hypertension and dyslipidaemia.
    - Hypertension may contribute to the risk of accelerated CKD progression.
    - Secondary causes of dyslipidaemia may include renal causes such as nephrotic syndrome (which typically presents with urinary albumin: creatinine ratio [ACR] levels greater than 2220 mg/g).
- Cardiovascular disease such as ischaemic heart disease, peripheral arterial disease, heart failure, and stroke disease.
- Renal anaemia (haemoglobin less than 11 g/dL [110 g/L]).
    - This may present with symptoms such as tiredness, shortness of breath, lethargy, and palpitations.
    - It may be due to reduced production of erythropoietin by the kidney, reduced red blood cell survival, and iron deficiency.

- Renal mineral and bone disorder.
    - This may present with bone pain, increased bone fragility, or extra-skeletal calcification, such as in the skin or blood vessels.
    - It is caused by disturbed vitamin D, calcium, parathyroid hormone (PTH), and phosphate metabolism due to impaired regulation of intestinal absorption and renal tubular excretion. This subsequently causes abnormalities in bone turnover and mineralisation. Serum calcium may be low or high initially, with vitamin D deficiency, raised serum phosphate, low serum calcium, and secondary or tertiary hyperparathyroidism seen in progressive CKD.
- Peripheral neuropathy and myopathy.
    - This may present with paraesthesia, sleep disturbance, and restless legs syndrome.
- Malnutrition.
    - This may be seen in ESRD, due to poor dietary intake and hypoalbuminaemia.
- Malignancy.
    - People with ESRD may have an excess cancer risk, particularly affecting the renal tract and thyroid gland. The exact cause is not known, but risk factors may include exposure to immunosuppressive agents, and immune dysregulation caused by chronic uraemia.
- End-stage renal disease (ESRD).
    - People with CKD stages 4–5 may need renal replacement therapy (RRT) in the form of dialysis or kidney transplantation.
- All-cause mortality.
    - The risk of all-cause mortality increases with progressive CKD.

Prognosis:

Decreasing estimated glomerular filtration rate (eGFR) and/or increasing proteinuria are independently associated with a poorer prognosis of chronic kidney disease (CKD).

Diagnosis:

Arrange investigations to assess for chronic kidney disease (CKD) in people with:

- Risk factors for CKD.
- An incidental finding of:
    - Raised serum creatinine and/or a serum estimated glomerular filtration rate (eGFR) of less than 60 mL/min/1.73 m2.
    - Proteinuria (a urinary albumin:creatinine ratio [ACR] of more than 30mg/g).
    - Persistent haematuria (two out of three urine dipstick tests show 1+ or more of blood), after exclusion of a urinary tract infection (UTI).
    - Urine sediment abnormalities, such as red blood cells (may indicate glomerular disease); white blood cells (may indicate pyelonephritis or interstitial nephritis); or

granular casts and renal tubular epithelial cells (seen in many parenchymal diseases).

- Possible clinical features of CKD. Be aware that disease is often asymptomatic in the early stages.

Assessment:

If a diagnosis of chronic kidney disease (CKD) is suspected, assess the person for possible associated clinical features.

- **Ask about:**
  - General symptoms, such as lethargy, itch, breathlessness, cramps (often worse at night), sleep disturbance, bone pain, or loss of appetite, vomiting, weight loss, and taste disturbance (often present with end-stage disease).
  - Urine output, such as polyuria (tubular concentrating ability is impaired); oliguria; nocturia (due to impaired solute diuresis or oedema); or anuria (due to possible acute kidney injury [AKI], obstructive uropathy causing urinary retention; or end-stage renal disease).
  - Any potentially nephrotoxic drugs, including over-the-counter or herbal medicines.
  - Any known risk factors for CKD or previous history of AKI.
  - Any associated co-morbidities or complications of CKD.
  - Any family history, such as autosomal dominant polycystic kidney disease.
  - Any associated clinical features of anxiety or depression.
- **Examine for signs of progressive CKD, such as:**
  - Uraemic odour (ammonia-like smell of the breath, may be present in advanced disease).
  - Pallor (due to renal anaemia).
  - Cachexia and signs of malnutrition.
  - Cognitive impairment (language, orientation, and attention may be particularly affected).
  - Dehydration or hypovolaemia (risk of AKI).
  - Tachypnoea (may be due to fluid overload, anaemia, or co-morbid ischaemic heart disease).
  - Hypertension (may be primary or secondary to CKD itself).
  - Palpable bilateral flank masses with possible hepatomegaly (suggests polycystic kidney disease with possible liver cysts).
  - Palpable distended bladder (suggests obstructive uropathy).
  - Peripheral oedema (may be due to renal sodium retention, hypoalbuminaemia, or co-morbid heart failure).
  - Peripheral neuropathy (may present with paraesthesia, sleep disturbance, and restless legs syndrome) or myopathy.

    - Frothy urine (may indicate proteinuria).
- Arrange investigations, as appropriate.
- Note: unexplained chronic renal impairment is an HIV indicator condition.

Initial investigations:

If a diagnosis of chronic kidney disease (CKD) is suspected, arrange initial investigations in primary care.

- Arrange blood tests for serum creatinine and estimated glomerular filtration rate (eGFR), and advise the person not to eat meat for at least 12 hours before the test.
    - If the eGFR is less than 60 mL/min/1.73 m2, repeat the test within two weeks (unless the eGFR is stable).
    - If the eGFR remains less than 60 mL/min/1.73 m2 on repeat, with no evidence of sudden deterioration in renal function suggesting acute kidney injury, repeat the eGFR within 3 months.
        - Note: interpret the eGFR result with caution if the person has extremes of muscle mass, is pregnant, has oedema, is malnourished or uses protein supplements, or is Asian or Chinese in origin.
- Arrange an early morning urine sample to measure the urinary albumin: creatinine ratio (ACR). If the result is:
    - Less than 30 mg/g(no proteinuria), no action is needed.
    - Between 30 and 700 mg/g, repeat the test within 3 months.
    - 700 mg/g or more, a repeat test is not needed as this indicates significant proteinuria.
    - Note: If the ACR is 700 mg/g or more, protein: creatinine ratio (PCR) can be used as an alternative to ACR.
    - Note: do not use a urine dipstick to test for proteinuria unless it can measure albumin at low levels and expresses the result as an albumin: creatinine ratio (ACR).
    - Note: If unexplained proteinuria is an incidental finding on a reagent strip, offer testing for CKD using eGFR creatinine and ACR.
    - Note: transient increases in urine ACR may be seen with menstruation, urinary tract infection (UTI), strenuous exercise, and upright posture ('orthostatic proteinuria').
- Arrange a urine dipstick test to check for haematuria.
    - If there is 1+ or more of blood on dipstick, arrange a mid-stream urine sample (MSU) to exclude a UTI, and manage accordingly.
    - If there is isolated persistent haematuria (two out of three urine dipstick tests show 1+ or more of blood after exclusion of a UTI), with no decrease in eGFR and no proteinuria.
- If the eGFR and urine ACR tests are repeated within 3 months:
    - Make a diagnosis of CKD if there is a persistent reduction in renal function (eGFR is less than 60 mL/min/1.73 m2) and/or proteinuria (urinary ACR is greater than 30 mg/g) lasting for at least three months.

        - Classify the category of CKD to determine the risk of disease progression and ongoing management.
        - Be aware that a significant increase in serum creatinine, for example by more than 20%, may indicate significant renal impairment, in the presence of normal eGFR readings (eGFR greater than 90 mL min/1.73 m2).
    - A diagnosis of CKD can be excluded if the eGFR is persistently greater than 60 mL/min/1.73 m2, and/or the urinary ACR is persistently less than 30 mg/g, and there are no other markers of kidney damage.
        - Arrange for serum creatinine, eGFR, urinary ACR, and urine dipstick testing to be checked annually if the person has not been diagnosed with CKD but has ongoing risk factors for CKD.
- Check the person's nutritional status, body mass index (BMI), blood pressure, and serum HbA1c and lipid profile to assess for cardiovascular risk factors.
- Consider arranging a renal tract ultrasound if indicated, for example if the person has suspected urinary tract stones or obstruction, or a family history of polycystic kidney disease and is aged over 20 years.

Classification of CKD

Chronic kidney disease (CKD) is classified using a combination of estimated glomerular filtration rate (eGFR) and urinary albumin: creatinine ratio (ACR).

- Increased ACR is associated with increased risk of adverse outcomes.
- Decreased GFR is associated with increased risk of adverse outcomes.
- Increased ACR and decreased GFR in combination multiply the risk of adverse outcomes.

Table 1. Classification of CKD using eGFR and urinary ACR categories.

eGFR (mL/min/1.73 m2)		Urinary ACR categories (mg/mmol)		
		< 30 Normal to mildly increased	30–300 Moderately increased	> 300 Severely increased
		A1	A2	A3
>=90 Normal and high	G1 (stage 1)	Not CKD in the absence of markers of kidney damage†	G1 A2	G1 A3
60–89 Mild reduction related to normal range for a young adult	G2 (stage 2)		G2 A2	G2 A3

45–59 Mild to moderate reduction	G3a (stage 3a)	G3a A1	G3a A2	G3a A3
30-44 Moderate to severe reduction	G3b (stage 3b)	G3b A1	G3b A2	G3b A3
15–29 Severe reduction	G4 (stage 4)	G4 A1	G4 A2	G4 A3
<15 Kidney failure	G5 (stage 5)	G5 A1	G5 A2	G5 A3
† Markers of kidney damage include albuminuria (urinary ACR more than 30mg/g), urine sediment abnormalities, electrolyte and other abnormalities due to tubular disorders, abnormalities detected by histology, structural abnormalities detected by imaging, or a history of kidney transplantation.				

Management:
From age 18 years onwards.
How should I monitor and identify disease progression?

If a person has a confirmed diagnosis of chronic kidney disease (CKD), arrange monitoring for disease progression and associated complications, and arrange specialist referral if appropriate.

- Identify any underlying cause and risk factors for disease progression which will influence the frequency of monitoring.
- Monitor renal function by checking serum creatinine and estimated glomerular filtration rate (eGFR) together with urinary albumin:creatinine ratio (ACR) to identify 'accelerated progression' of CKD.
    - To assess the rate of progression, repeat the serum eGFR three times over at least 3 months.
        - If present, assess for any reversible cause (such as potentially nephrotoxic drugs or volume depletion), arrange a renal tract ultrasound scan to identify an underlying structural cause, and arrange referral to a nephrology specialist for further assessment and management.
    - If the person has, or is at risk of, acute kidney injury (AKI):
        - Consider stopping any potentially nephrotoxic drugs.

        - Monitor for the development or progression of CKD for at least 2–3 years after an episode of AKI, even if serum creatinine has returned to baseline.
- Take into account the person's baseline eGFR and likely trajectory if the current rate of eGFR decline was to continue, to help decide on management and whether specialist referral is needed.
    - Note previous trends of eGFR and ACR results, as the course of CKD is often non-linear. More frequent monitoring may be needed in people with previous variable or erratic renal function, and less frequent monitoring may be needed for those with stable results.
    - Table 1 provides a guide to recommended monitoring frequency.

Table 1. Frequency of eGFR monitoring (number of times per year) for people with or at risk of CKD.

Serum eGFR (mL/min/1.73 m2)	Urinary ACR (mg/g)		
	A1 (< 30) Normal to mildly increased	A2 (30–300) Moderately increased	A3 (> 300) Severely increased
G1 >=90 Normal and high	=< 1	1	>= 1
G2 60–89 Mild reduction related to normal range for a young adult	=< 1	1	>= 1
G3a 45–59 Mild to moderate reduction	1	1	2
G3b 30–44 Moderate to severe reduction	=< 2	2	>= 2
G4 15–29 Severe reduction	2	2	3
G5 <15 End-stage kidney failure	4	>=4	>=4

- Arrange a full blood count (FBC) to exclude renal anemia for people with CKD category stages 3b, 4, and 5, or if a person develops symptoms suggestive of anaemia.
    - The frequency of subsequent monitoring depends on the FBC result and clinical judgement.

    - Consider checking FBC in people with CKD stages 1, 2, and 3a if anaemia is suspected clinically.
    - If anaemia is present, arrange tests to exclude other causes of anaemia, such as iron deficiency.
    - If renal anaemia is suspected, arrange referral to a nephrology specialist for further assessment and management.
- Arrange serum calcium, phosphate, vitamin D, and parathyroid hormone tests to exclude renal metabolic and bone disorder for people with CKD category stages 4 or 5. The frequency of subsequent monitoring depends on the results and clinical judgement.
    - Consider checking serum parathyroid hormone and vitamin D level in people with CKD stages 1, 2, 3a, and 3b if bone disease is suspected clinically, for example, if there is unexplained hypo- or hypercalcaemia

Who should I refer?

- Arrange an urgent 2-week wait referral if there is isolated persistent hematurea and a urological cancer is suspected.
- Arrange referral to a nephrology specialist, the urgency depending on clinical judgement, if there is:
    - A 5-year risk of needing renal replacement therapy of greater than 5% (measured using the 4-variable Kidney Failure Risk Equation).
    - An estimated glomerular filtration rate (eGFR) of less than 30 mL/min/1.73 m2.
        - Disease monitoring may take place in primary care as part of a shared care agreement.
    - Accelerated progression of chronic kidney disease (CKD), defined as:
        - A sustained decrease in eGFR of 15 mL/min/1.73 m2 or more within 12 months.
    - A urinary albumin:creatinine ratio (ACR) of 700 mg/g or more, unless proteinuria is known to be associated with diabetes mellitus and is managed appropriately.
    - A urinary ACR of 300 mg/g or more together with persistent haematuria, after exclusion of a urinary tract infection (UTI).
    - Hypertension that remains uncontrolled despite the use of at least four antihypertensive drugs at therapeutic doses.
    - A suspected or confirmed rare or genetic cause of CKD, such as polycystic kidney disease.
    - Suspected renal artery stenosis.
        - This should be suspected if there is a greater than 15 mL/min/1.73 m2 reduction in eGFR within 3 months of starting (or increasing the dose of) a renin-angiotensin system antagonist, refractory hypertension, pulmonary oedema, and/or a renal artery bruit.
    - A suspected complication of CKD. Specialist management of:

        - A decline in nutritional status or malnutrition, and persistent hyperkalaemia, may include specialist dietary advice about potassium, phosphate, calorie and salt intake; and end-stage renal disease (ESRD) may require renal replacement therapy (haemodialysis/peritoneal dialysis or eventual kidney transplantation).
        - Renal anaemia may include treatment with erythropoiesis-stimulating agents, and possible intravenous iron if there is associated refractory iron deficiency.
        - Renal mineral and bone disorder may involve phosphate dietary restriction, use of phosphate binders, and specialist endocrinology input.
        - Persistent metabolic acidosis may involve oral bicarbonate supplementation.
- Arrange referral to a urology specialist, the urgency depending on clinical judgement, if there is:
    - Suspected urinary tract obstruction (for example the bladder is palpable or hydronephrosis is seen on renal tract ultrasound).
    - Note: emergency admission may be required if the person is in urinary retention; has severe hyperkalaemia (potassium greater than 6 mmol/L); severe uraemia; or signs of fluid overload or dehydration.

How should I manage a person in primary care?

If a person has a confirmed diagnosis of chronic kidney disease (CKD), arrange regular follow-up in primary care, the frequency depending on clinical judgement.

- **Assess for hypertension. If the person does not have diabetes and has:**
    - A urinary albumin:creatinine ratio (ACR) of less than 30 mg/g and associated hypertension, manage as for adults without CKD.
    - A urinary ACR of 30 mg/g or more and associated hypertension, prescribe a low-cost renin-angiotensin system antagonist (such as lisinopril or losartan) to achieve the optimal tolerated dose, depending on contraindications, cautions, and drug interactions.
        - Do not prescribe a combination of renin-angiotensin system antagonists to people with CKD, such as lisinopril and losartan.
    - Hypertension which remains uncontrolled despite the use of at least four antihypertensive drugs at therapeutic doses, arrange referral to a nephrology specialist.
- **Consider prescribing dapagliflozin or empagliflozin for CKD if:**
    - It is an add-on to optimised standard care including the highest tolerated licensed dose of angiotensin-converting enzyme (ACE) inhibitors or angiotensin-II receptor antagonists (AIIRAs), unless these are contraindicated, and people have an estimated glomerular filtration rate (eGFR) of 20 ml/min/1.73 m2 to 75 ml/min/1.73 m2 at the start of treatment and have:
        - Type 2 diabetes or
        - microalbuminurea

- **Prescribe an antiplatelet drug for the secondary prevention of CVD.**
- **Ensure the person is offered immunizations for influenza and pneumococcal disease to reduce infection risk, if appropriate.**

What self-management advice should I give?

If a person has a confirmed diagnosis of chronic kidney disease (CKD), offer self-management advice in relation to the underlying cause, disease severity, associated complications, and risk of progression.

- Advise on healthy lifestyle measures, such as:
  - Stopping smoking if appropriate.
  - Drinking alcohol in moderation.
  - Maintaining a healthy body weight.
  - Eating a healthy diet and taking regular exercise.
    - A low-protein diet (dietary protein intake of less than 0.8/kg/day) is not routinely recommended.
    - Specialist dietary advice about potassium, phosphate, calorie, and salt intake may be arranged by the person's specialist for people with end-stage CKD following referral.
- Advise the person to avoid the use of over-the-counter nonsteroidal anti-inflammatory drugs (NSAIDs) where possible and avoid herbal remedies, and use protein supplements with caution.
- Advise on the increased risk of acute kidney injury (AKI) if there is severe, intercurrent illness.

Table 1. Low starting, usual starting, maintenance, and maximum doses of ACE- inhibitors used for hypertension.

ACE-inhibitor	Low starting doses*	Usual starting dose	Usual maintenance dose	Maximum dose
Captopril	6.25 mg to 12.5 mg single dose, then 6.25 to 12.5 mg twice a day	12.5 to 25 mg twice a day	50 to 75 mg twice a day	75 mg twice a day
Enalapril	2.5 mg once a day	5 mg once a day	20 mg once a day	40 mg once a day
Fosinopril	10 mg once a day	10 mg once a day	10 to 40 mg once a day	40 mg once a day

Imidapril	2.5 mg once a day	5 mg once a day	10 mg once a day	20 mg once a day
Lisinopril	2.5 mg once a day	10 mg once a day	20 mg once a day	80 mg once a day
Perindopril arginine	2.5 mg once a day	5 mg once a day	5 to 10 mg once a day	10 mg once a day
Quinapril	2.5 mg once a day	10 mg once a day	10 to 20 mg twice daily	40 mg twice daily
Ramipril	1.25 mg once a day	1.25 to 2.5 mg once a day	2.5 mg to 10 mg once a day	10 mg once a day
Trandolapril	500 micrograms once a day	500 micrograms once a day	1 to 2 mg once a day	4 mg once a day
* Lower starting doses are required for people who are more prone to the adverse effects of ACE-inhibitors (such as the elderly, frail, or people taking diuretics).				

Table 2. Low starting, usual starting, maintenance, and maximum doses of AIIRAs used for hypertension.

ARIIA	Low starting doses*	Usual starting dose	Usual maintenance dose	Maximum dose
Azilsartan	20 to 40 mg once a day	40 mg once a day	40 to 80 mg once a day	80 mg once a day
Candesartan	4 mg once a day	8 mg once a day	8 mg once a day	32 mg once a day
Eprosartan	600 mg once a day	600 mg once a day	600 mg once a day	600 mg once a day
Irbesartan	75 mg once a day	150 mg once a day	150 mg once a day	300 mg once a day
Losartan	25 mg once a day	50 mg once a day	50 mg once a day	100 mg once a day

Olmesartan	10 mg once a day	10 mg once a day	20 mg once a day	40 mg once a day
Telmisartan	20 mg once a day	20 to 40 mg once a day	40 mg once a day	80 mg once a day
Valsartan	40 mg once a day	80 mg once a day	80 to 160 mg once a day	320 mg once a day
* Lower starting doses are required for people who are more prone to the adverse effects of AIIRAs (such as the elderly, frail, or people taking diuretics).				

Topic: Chronic kidney disease and anemia

Patients with chronic kidney disease (CKD) may develop anemia due to a variety of factors, the most significant of which is reduced erythropoietin levels.

This is usually a normochromic normocytic anemia and becomes apparent when the GFR is less than 35 ml/min (other causes of anemia should be considered if the GFR is > 60 ml/min).

Anemia in CKD predisposes to the development of left ventricular hypertrophy - associated with a threefold increase in mortality in renal patients

Causes of anemia in renal failure:

- Reduced erythropoietin levels - the most significant factor.
- Reduced erythropoiesis due to toxic effects of uremia on bone marrow.
- Reduced absorption of iron.
- Anorexia/nausea due to uremia.
- Reduced red cell survival (especially in hemodialysis).
- Blood loss due to capillary fragility and poor platelet function.
- Stress ulceration leading to chronic blood loss.

Management:

- The 2011 NICE guidelines suggest a target hemoglobin of 10 - 12 g/dl
- Determination and optimization of iron status should be carried out prior to the administration of erythropoiesis-stimulating agents (ESA). Many patients, especially those on hemodialysis, will require IV iron
- ESAs such as erythropoietin and darbepoetin should be used in those 'who are likely to benefit in terms of quality of life and physical function'

Topic: Chronic kidney disease and mineral bone disease management

Pathophysiology

Basic problems in chronic kidney disease (CKD):

- 1-alpha hydroxylation normally occurs in the kidneys → CKD leads to low vitamin D
- the kidneys normally excrete phosphate → CKD leads to high phosphate

This, in turn, causes other problems:

- The high phosphate level 'drags' calcium from the bones, resulting in osteomalacia
- low calcium: due to lack of vitamin D, high phosphate
- secondary hyperparathyroidism: due to low calcium, high phosphate and low vitamin D

Management

The aim is to reduce phosphate and parathyroid hormone levels.

Overview

- reduced dietary intake of phosphate is the first-line management
- phosphate binders
- vitamin D: alfacalcidol, calcitriol
- parathyroidectomy may be needed in some cases

Phosphate binders:

- Aluminum-based binders are less commonly used now
- Calcium-based binders:
    - Problems include hypercalcemia and vascular calcification
- Sevelamer:
    - a non-calcium-based binder that is now increasingly used
    - binds to dietary phosphate and prevents its absorption
    - also appears to have other beneficial effects including reducing uric acid levels and improving the lipid profiles of patients with chronic kidney disease.

Topic: Hematuria

The management of patients with hematuria is often difficult due to the absence of widely followed guidelines. It is sometimes unclear whether patients are best managed in primary care, by urologists or by nephrologists.

The terminology surrounding hematuria is changing. Microscopic or dipstick positive hematuria is increasingly termed non-visible hematuria whilst macroscopic hematuria is termed visible hematuria.

Non-visible hematuria is found in around 2.5% of the population.

Causes of transient or spurious non-visible hematuria

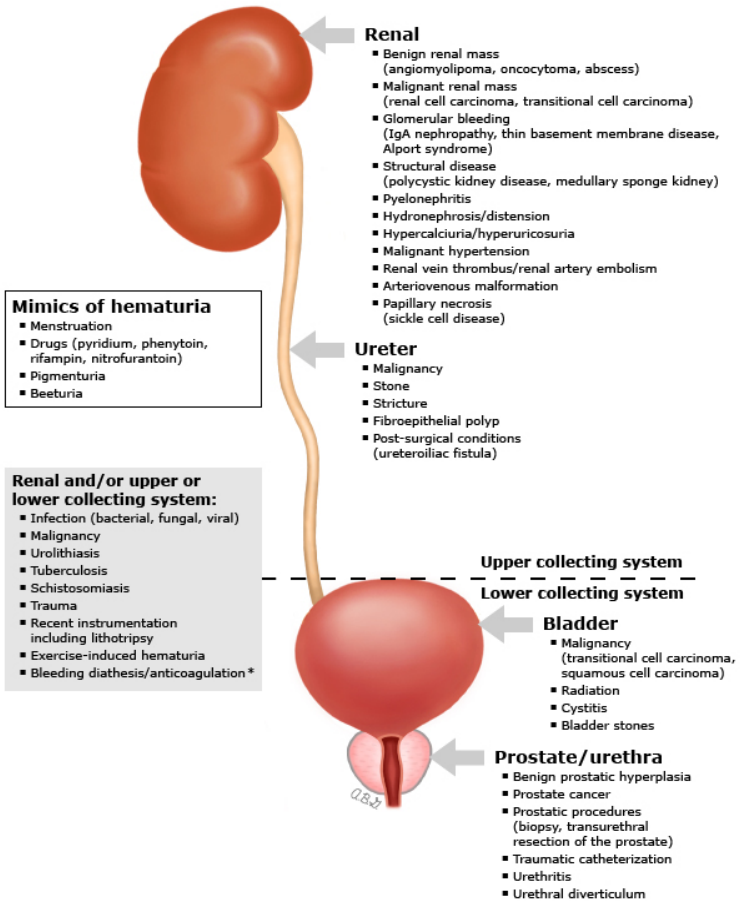
- Urinary tract infection
- Menstruation
- Vigorous exercise (this normally settles after around 3 days)
- Sexual intercourse

Causes of persistent non-visible hematuria

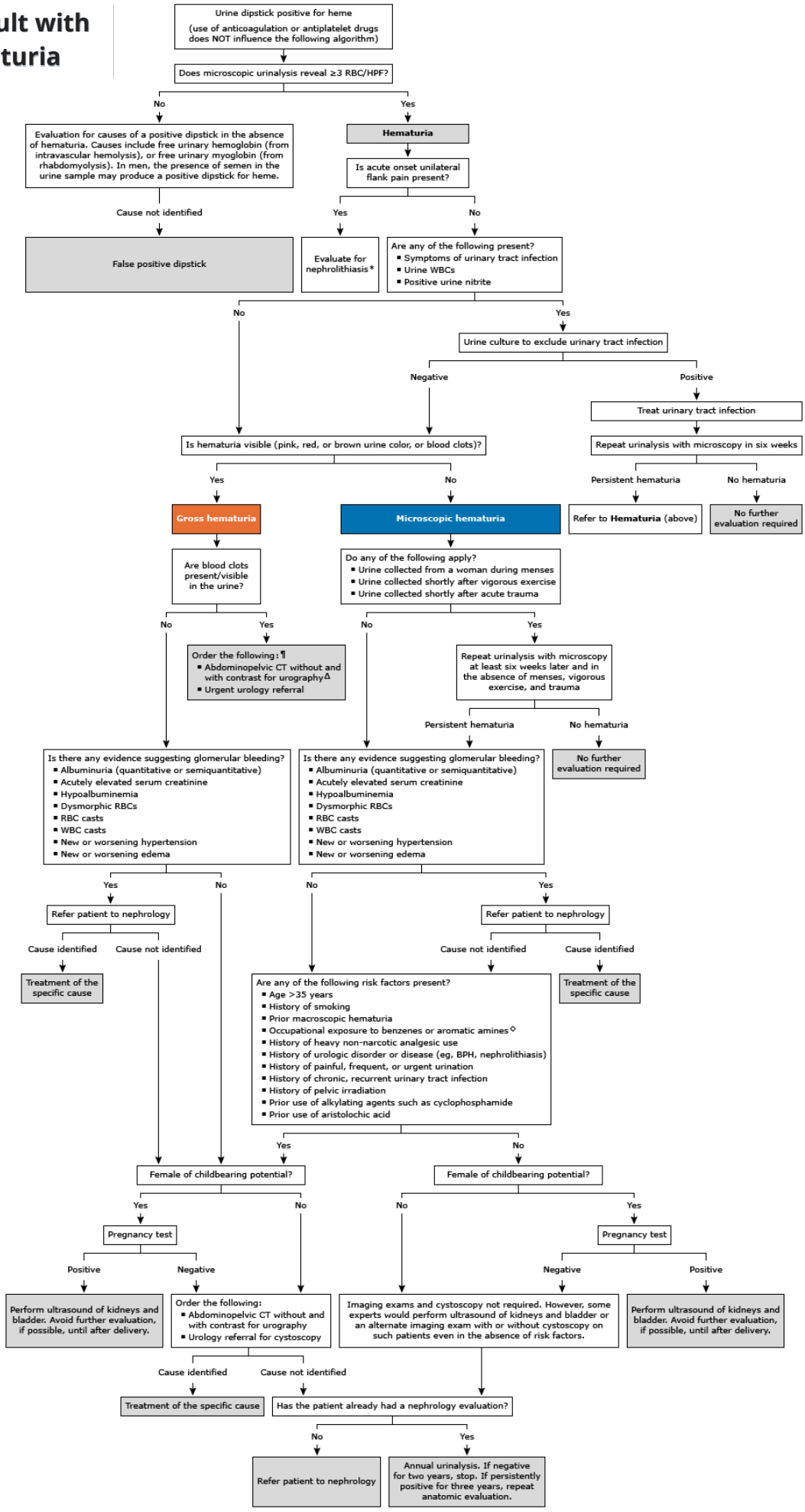
- cancer (bladder, renal, prostate)
- stones
- benign prostatic hyperplasia
- prostatitis
- urethritis e.g., Chlamydia
- renal causes: IgA nephropathy, thin basement membrane disease

Spurious causes - red/orange urine, where blood is not present on dipstick

- foods: beetroot, rhubarb
- drugs: rifampicin, doxorubicin



Evaluation of the adult with asymptomatic hematuria



Haematuria (PassMedicine)

Management

- Current evidence does not support screening for haematuria. The incidence of non-visible haematuria is similar in patients taking aspirin/warfarin to the general population hence these patients should also be investigated.

- Testing
  - urine dipstick is the test of choice for detecting haematuria
  - persistent non-visible haematuria is often defined as blood being present in 2 out of 3 samples tested 2-3 weeks apart
  - renal function, albumin:creatinine (ACR) or protein:creatinine ratio (PCR) and blood pressure should also be checked
  - urine microscopy may be used but time to analysis significantly affects the number of red blood cells detected

Topic: IgA vasculitis (Henoch-Schönlein purpura)

Henoch-Schonlein purpura (HSP) is an IgA mediated small vessel vasculitis. There is a degree of overlap with IgA nephropathy (Berger's disease). HSP is usually seen in children following an infection.
In contrast to many other forms of systemic vasculitis, IgAV is typically self-limited. The disease is characterized by a tetrad of clinical manifestations:

1. **Palpable purpura in patients with neither thrombocytopenia nor coagulopathy**
2. **Arthritis/arthralgia, abdominal pain, kidney disease (hematurea)**

Diagnosis:

- The diagnosis of IgAV is usually based upon clinical manifestations of the disease.
- There are no diagnostic laboratory tests for IgAV.
- In patients with an incomplete or unusual presentation, biopsy of the affected organ (eg, skin or kidney) demonstrating predominantly IgA deposition supports the diagnosis of IgAV.

Investigations:

- Patients in whom the diagnosis is at all in doubt should have a CBC, prothrombin time (PT), and urinalysis.
- The presence of thrombocytopenia or a coagulopathy largely excludes the diagnosis of IgAV.
- A serum creatinine level should be obtained in children with hypertension or an abnormal urinalysis.
- Serum creatinine should be assayed in all adult patients with IgAV because of the increased risk of significant kidney disease.
- Imaging studies are performed in those with significant gastrointestinal symptoms.
- Abdominal ultrasonography is indicated in patients with severe abdominal pain. It can detect increased bowel wall thickness, hematomas, peritoneal fluid, and intussusception.
- Contrast studies may miss intussusception in patients with IgAV.

Treatment:

- IgAV is a self-limited disease
- General management: Most patients with immunoglobulin A vasculitis (IgAV), formerly called Henoch-Schönlein purpura (HSP), may be cared for in the ambulatory setting with therapy directed toward adequate oral hydration, bed rest, and symptomatic relief of joint and abdominal pain.
- Treatment of joint and/or abdominal pain: In patients with joint and/or abdominal pain, UpToDate suggest the use of a nonsteroidal anti-inflammatory agent (Grade 2C). We typically use naproxen (10 to 20 mg/kg divided into two doses per day). A maximum dose of naproxen of 1500 mg per day may be used for a few days provided that adequate hydration is maintained. If more than a week of nonsteroidal anti-inflammatory drug (NSAID) treatment is necessary, the dose of naproxen should not exceed 1000 mg per day.
- In patients with severe abdominal pain that interferes with their oral intake and who fail to respond to a nonsteroidal anti-inflammatory agent, **oral prednisone** is suggested (1 to 2 mg/kg per day) for children or adults, with a maximum dose of 60 to 80 mg per day.
- Treatment of nephropathy is generally supportive, bp and urine test should be monitored.
- Empiric prophylaxis with corticoid administration to prevent kidney or gastrointestinal complications **is not recommended.**

Prognosis:

- usually excellent, HSP is a self-limiting condition, especially in children without renal involvement, around one-third of patients have a relapse.

Topic: IgA nephropathy (IgAN)

IgA nephropathy (also known as Berger's disease) is the commonest cause of glomerulonephritis worldwide. It classically presents as macroscopic hematuria in young people following an upper respiratory tract infection.

Clinical features:

- Mostly: young male, recurrent episodes of macroscopic hematuria, typically associated with a recent respiratory tract infection
- Most patients with IgAN present with either gross hematuria (single or recurrent), usually accompanying an upper respiratory infection, or microscopic hematuria with or without mild proteinuria incidentally detected on a routine examination.
- These symptoms are often accompanied by hypertension.
- Less commonly, patients may present with either nephrotic syndrome or an acute, rapidly progressive glomerulonephritis. Patients may rarely develop acute kidney injury with or without oliguria, due to crescentic IgAN or to gross hematuria causing tubular occlusion and/or damage by red cells.

Associated conditions:

- Chronic liver disease, celiac disease, HIV infection
- Monoclonal gammopathy of renal significance (MGRS).
- In addition, IgAN may rarely be seen with other glomerular diseases, including minimal change disease, membranous nephropathy, and granulomatosis with polyangiitis.

Diagnosis:

- The diagnosis of IgAN should be suspected in any patient who presents with:
  - one or more episodes of gross hematuria (especially if accompanied by an upper respiratory infection).
  - Persistent microscopic hematuria with or without proteinuria.
  - Slowly progressive kidney function impairment.
- The diagnosis is confirmed by kidney biopsy. However, a kidney biopsy is usually performed only if there are signs suggestive of more severe or progressive disease, such as persistent proteinuria of at least 500 mg per day or an elevated serum creatinine concentration.

Differentiating between IgA nephropathy and post-streptococcal glomerulonephritis:

- Post-streptococcal glomerulonephritis is associated with low complement levels
- Main symptom in post-streptococcal glomerulonephritis is proteinuria (although haematuria can occur)
- There is typically an interval between URTI and the onset of renal problems in post-streptococcal glomerulonephritis

Management:

- Isolated hematuria, no or minimal proteinuria (less than 500 to 1000 mg/day), and a normal glomerular filtration rate (GFR)
  - no treatment needed, other than follow-up to check renal function
- Persistent proteinuria (above 500 to 1000 mg/day), a normal or only slightly reduced GFR
  - initial treatment is with ACE inhibitors
- If there is active disease (e.g., falling GFR) or failure to respond to ACE inhibitors
  - immunosuppression with corticosteroids

Prognosis:

- 25% of patients develop ESRF
- markers of good prognosis: frank haematuria
- markers of poor prognosis: male gender, proteinuria (especially > 2 g/day), hypertension, smoking, hyperlipidaemia, ACE genotype DD

Topic: Post-streptococcal glomerulonephritis (PSGN)

- Post-streptococcal glomerulonephritis typically occurs 7-14 days following a group A beta-haemolytic Streptococcus infection (usually Streptococcus pyogenes).
- It is caused by immune complex (IgG, IgM and C3) deposition in the glomeruli.
- Poststreptococcal glomerular nephritis (PSGN) is the most common cause of acute nephritis worldwide.
- It primarily occurs in the developing world.
- The risk of PSGN is greatest in children between 5 and 12 years of age, and in older adults greater than 60 years of age.

Features

- General: headache, malaise, visible hematurea
- Proteinuria: this may result in edema, HTN, oliguria

Investigations:

Bloods:

- Abnormal urinalysis (dysmorphic red blood cells, varying degrees of proteinuria, red blood cell casts, and pyuria)
- low C3, raised ASO titer

Diagnosis:

PSGN is typically diagnosed based upon the findings of:

Acute nephritis (eg, hematuria with or without red blood cell casts and proteinuria)

AND

demonstration of a recent GAS infection (eg, positive throat or skin culture or serologic tests [anti-streptolysin [ASO] or streptozyme test]).

Renal biopsy features:

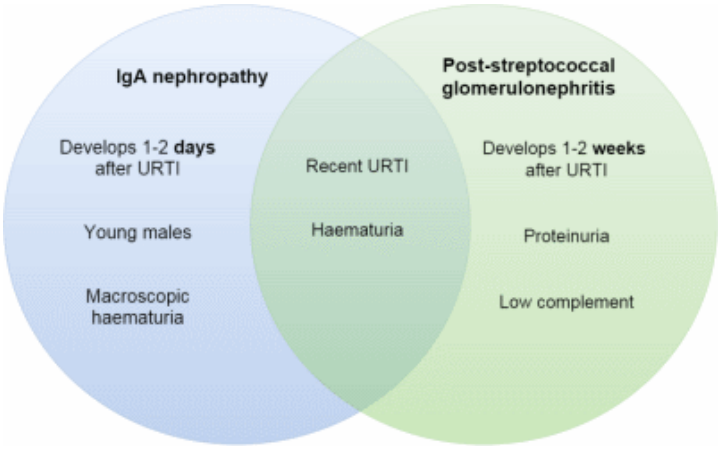
- post-streptococcal glomerulonephritis causes acute, diffuse proliferative glomerulonephritis
- endothelial proliferation with neutrophils
- electron microscopy: subepithelial 'humps' caused by lumpy immune complex deposits
- immunofluorescence: granular or 'starry sky' appearance

Prognosis:

Most patients, particularly children, have complete clinical recovery, and resolution of their disease process begins within the first two weeks.

Management:

- There is no specific therapy to treat PSGN.
- Management is supportive and is focused upon treating the volume overload that causes the clinical complications of PSGN.
- These general measures include sodium and water restriction, and diuretic therapy



Topic: Minimal change disease (MCD)

Minimal change disease (MCD) is a major cause of the nephrotic syndrome in children (approximately 90 percent) and in a minority of adults (approximately 10 percent).

MCD and focal segmental glomerulosclerosis (FSGS) are both examples of pathogenic mechanisms that primarily affect the podocyte ("podocytopathies")

Etiology:

- Most cases of MCD are idiopathic (or primary) and are not clearly associated with an underlying disease or event.

- With secondary MCD, the onset of nephrotic syndrome occurs concurrently or following an extraglomerular or glomerular disease process. MCD is associated with a variety of secondary causes, including drugs, neoplasms, infections, and allergy.

Clinical features:

In most patients, MCD is characterized by the sudden onset over days to a week or two of the signs and symptoms of the nephrotic syndrome, often following an upper respiratory or systemic infection.

The typically sudden onset of nephrotic syndrome in MCD is in contrast to the gradual increase in proteinuria over weeks to months in most other causes of nephrotic syndrome.

Microscopic hematuria and acute kidney injury (AKI) may also be seen in adults with MCD.

Diagnosis:

- The diagnosis of MCD should be suspected in any adult patient presenting with symptoms and signs of the nephrotic syndrome, especially if the onset is relatively acute (over days to a week or two).
- A kidney biopsy is required to establish the diagnosis of MCD in adults and to exclude other causes of the nephrotic syndrome.
- There are no specific laboratory findings that can be used to distinguish MCD from other forms of nephrotic syndrome.

Treatment:

- The goal of therapy for patients with minimal change disease (MCD) is to significantly reduce proteinuria and thereby induce remission of the nephrotic syndrome.
- This is primarily achieved with the use of immunosuppressive agents, most commonly glucocorticoids.
    - oral corticosteroids: majority of cases (80%) are steroid-responsive
    - cyclophosphamide is the next step for steroid-resistant cases
- Untreated MCD is associated with potentially fatal complications, including sepsis and thrombosis.

Prognosis:

Is overall good, although relapse is common. Roughly:

- 1/3 have just one episode
- 1/3 have infrequent relapses
- 1/3 have frequent relapses which stop before adulthood

Topic: Nephrotic syndrome.

The nephrotic syndrome is defined by the presence of:

1. Heavy proteinuria (protein excretion greater than 3.5 g/24 hours in an adult)
2. hypoalbuminemia (less than 3 g/dL)
3. Peripheral edema.

Etiology:

- The predominant cause of the nephrotic syndrome in children is minimal change disease.
- Approximately 30 percent of adults with the nephrotic syndrome have a systemic disease such as:
  - Diabetes mellitus, amyloidosis, or systemic lupus erythematosus; the remaining cases are usually due to primary disorders including minimal change disease, focal segmental glomerulosclerosis (FSGS), and membranous nephropathy.
- Drugs
  - Gold (sodium aurothiomalate), Penicillamine.

Diagnosis:

- The etiology of heavy proteinuria may be suggested from the history and physical examination.
- In most adults, a kidney biopsy is required to establish the diagnosis.

Treatment:

Treatment includes the administration of:

- An angiotensin-converting enzyme (ACE) inhibitor or angiotensin receptor blockers (ARBs) to lower intraglomerular pressure, and dietary sodium restriction and loop diuretics to slowly reduce edema.
- The lipid abnormalities induced by the nephrotic syndrome usually reverse with resolution of the disease, but lipid-lowering therapy may be indicated in selected patients.
- Arterial and venous thromboembolism are typically treated with heparin followed by warfarin for as long as the patient remains nephrotic.
- Patients with primary (idiopathic) nephrotic syndrome often receive immunosuppressive therapy.

Topic: Renal papillary necrosis

Renal papillary necrosis describes the coagulative necrosis of the renal papillae due to a variety of causes.

Causes:

- Severe acute pyelonephritis
- Diabetic nephropathy
- Obstructive nephropathy
- Analgesic nephropathy
- Phenacetin was the classic cause but this has now been withdrawn
- NSAIDs
- Sickle cell anaemia

Clinical features:

The most common presenting symptoms in symptomatic patients include fever and chills, flank and/or abdominal pain, hematuria and Proteinuria.

Acute kidney injury with oliguria or anuria is rare; when these symptoms develop, the disease may be fulminant, requiring dialysis and potentially resulting in death.

Investigations:

- The most common urinalysis findings include proteinuria, pyuria, bacteriuria, and low urine specific gravity.
- Urinalysis (i.e., routine, microscopic)
- CBC
- Complete metabolic panel
- Prothrombin time and activated partial thromboplastin time determinations.

In febrile patients, obtain urine and blood cultures under sterile conditions.

- If the patient is prostrate and obtunded, measure arterial blood gases and perform standard electrocardiography and chest radiography.

Management:

General considerations:

- Patients with known or suspected renal papillary necrosis should limit or completely avoid the use of analgesics.
- Other nephrotoxic medications should also be avoided.
- The routine use of indwelling catheters should be discouraged except when clinically indicated.
- Clean intermittent catheterization, although more time-consuming, is equally effective for the purpose of monitoring output and is less likely to cause nosocomial urinary tract infection.

Medical Therapy:

Basically, treatment for renal papillary necrosis consists of the following:

- Ameliorate the ischemia with hydration and alkalinization.
- Treat the underlying cause of the renal papillary necrosis (e.g., maintain normal glycemic state)
- Institute targeted antibiotic therapy.

Surgical Therapy:

- Patients with renal papillary necrosis may require diagnostic and therapeutic urologic intervention. The urologist is responsible for evaluating any obstruction, hematuria, overwhelming infection, and associated malignancies and for preventing recurrences of these sequelae.
- The recommended treatment is to drain the dilated collecting system either endoscopically or percutaneously.

- Nephrectomy may be life-saving in patients with overwhelming infection (i.e., emphysematous pyelonephritis).

Topic: Diabetes insipidus

Diabetes insipidus (DI) is a condition characterized by either a decreased secretion of antidiuretic hormone (ADH) from the pituitary (cranial DI) or an insensitivity to antidiuretic hormone (nephrogenic DI).

Causes of cranial DI:

- idiopathic
- post head injury
- pituitary surgery
- craniopharyngiomas
- infiltrative
    - histiocytosis X
    - sarcoidosis
- DIDMOAD is the association of cranial Diabetes Insipidus, Diabetes Mellitus, Optic Atrophy and Deafness (also known as Wolfram's syndrome)
- Haemochromatosis.

Causes of nephrogenic DI:

- Genetic: the more common form affects the vasopressin (ADH) receptor, the less common form results from a mutation in the gene that encodes the aquaporin 2 channel
- Electrolytes: hypercalcemia, hypokalemia
- lithium
    - lithium desensitizes the kidney's ability to respond to ADH in the collecting ducts
- demeclocycline
- tubulo-interstitial disease: obstruction, sickle-cell, pyelonephritis.

Features:

- Polyuria
- Polydipsia

Investigation:

- High plasma osmolality, low urine osmolality
- a urine osmolality of >700 mOsm/kg excludes diabetes insipidus
- water deprivation test

Management:

- Nephrogenic diabetes insipidus: thiazides, low salt/protein diet.
- Central diabetes insipidus can be treated with desmopressin.

Topic: Spironolactone

Pharmacologic Category:

- Potassium Sparing; Mineralocorticoid (Aldosterone) Receptor Antagonist

Brand name:

- **Aldactone**

Uses and Doses:

For hypertension (Chronic):

- Tablet: Oral: Initial: 25 mg once daily; titrate as needed after ~2 to 4 weeks based on response and tolerability up to 100 mg once daily.
- Some experts recommend a starting dose of 12.5 mg once daily and generally do not exceed 50 mg once daily in the absence of primary aldosteronism.

Heart failure:

- Tablet: Oral: Initial: 12.5 to 25 mg once daily; may double the dose every 4 weeks if serum potassium remains <5 mEq/L and renal function is stable, up to a maximum target dose of 50 mg/day in 1 to 2 divided doses.

Ascites due to cirrhosis:

Generally used in combination with furosemide, but may be used as monotherapy for patients with hypokalemia.

Tablet: Oral: Initial: 100 mg once daily; titrate every 3 to 5 days based on response and tolerability; usual maximum dose: 400 mg once daily.

For small-volume ascites in patients who weigh ≤50 kg, some experts recommend a starting dose of 50 mg once daily.

Primary aldosteronism (Tablet only):

Tablet: Oral: Initial: 12.5 to 25 mg once daily; gradually titrate to the lowest effective dose; usual maximum dose: 400 mg/day.

For surgical candidates, the last dose should be administered the day of surgery; discontinue spironolactone on postoperative day 1**.**

Renal adjustment dose:

Ascites due to cirrhosis; hypertension; primary aldosteronism:

There are no specific dosage adjustments provided in the manufacturer's labeling (has not been studied); use with caution.

Heart failure:

- eGFR >50 mL/minute/1.73 m2: **No initial dosage adjustment** necessary.
- eGFR 30 to 50 mL/minute/1.73 m2: **Initial: 12.5 mg once daily** or every other day; may double the dose every 4 weeks if serum potassium remains <5 mEq/L and renal function is stable, up to a maximum target dose of 25 mg/day.
- eGFR <30 mL/minute/1.73 m2: **Use not recommended**; heart failure clinical trials excluded patients with serum creatinine ≥2.5 mg/dL

Administration:

Oral: Tablet:

- Administer with or without food; however, administer consistently with respect to food.
- Suspension: Shake well before administering dose. Administer with or without food; however, administer consistently with respect to food.

Adverse Reactions (Significant): Considerations

1. **Gynecomastia:** Usually Bilateral, reversible following discontinuation of therapy.
2. **Hyperkalemia**
3. hyperuricemia
4. Gastrointestinal: Abdominal cramps, diarrhea, gastritis, gastrointestinal hemorrhage, gastrointestinal ulcer, nausea, vomiting.
5. Hypersensitivity: Anaphylaxis

Contraindications:

- Hyperkalemia; Addison disease; concomitant use with eplerenone.
- Canadian labelling contraindications (not in US labelling):
  - Breastfeeding
  - Pregnancy
  - Acute renal insufficiency
  - Severe renal impairment (eGFR <30 mL/minute/1.73 m2)
- Patients taking spironolactone for primary aldosteronism (PA) who are planning to become pregnant should be switched to other agents prior to conception when possible (Forestiero 2022).
- Spironolactone is considered compatible with breastfeeding (WHO 2002), According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother.

Special populations:

Older adult: Avoid use of tablets >25 mg/day in older adult patients with heart failure

Drug Interactions:

- Alfuzosin: May enhance the hypotensive effect of Blood Pressure Lowering Agents.
- Aspirin: May diminish the therapeutic effect of Spironolactone.
- Atorvastatin: May enhance the adverse/toxic effect of Spironolactone.
- Ciprofloxacin (Systemic): Spironolactone may enhance the arrhythmogenic effect of Ciprofloxacin.
- Angiotensin II Receptor Blockers: May enhance the hyperkalemic effect of Potassium-Sparing Diuretics.
- Angiotensin-Converting Enzyme Inhibitors: Potassium-Sparing Diuretics may enhance the hyperkalemic effect of Angiotensin-Converting Enzyme Inhibitors.

Pharmacokinetics:

- Duration: Tablet: 2 to 3 days